**Date:** October 15, 2026

**To:** Professor Vladimir Zwass, Editor-in-Chief
*Journal of Management Information Systems* (Taylor & Francis)

**Subject:** Manuscript Submission: *"The Epistemic Fluency Trap in AI-Augmented Security Operations: Generative Rationale Fluency, Contested Evidence Cues, and Analyst Verification"*

Dear Professor Zwass and the Editorial Board of *Journal of Management Information Systems*,

We are pleased to submit our original research manuscript entitled **"The Epistemic Fluency Trap in AI-Augmented Security Operations: Generative Rationale Fluency, Contested Evidence Cues, and Analyst Verification"** for consideration for publication in *Journal of Management Information Systems* (*JMIS*).

## Why This Manuscript Fits *JMIS*

Enterprise Security Operations Centers (SOCs) increasingly deploy generative AI copilots and machine-learning triage pipelines as organizational decision-support systems. While prior Information Systems (IS) delegation and explainable AI (XAI) research assumes that linguistically coherent explanations improve human sensemaking and metaknowledge, SOC telemetry contains attacker-controllable string fields (`CommandLine`, `ProcessCommandLine`, and alert payload strings) that expose AI copilots and text-inclusive classifiers to **Indirect Telemetry Prompt Injection (ITPI)**. This manuscript addresses the core themes of *JMIS* across four tightly integrated layers:

1. **Behavioral and Decision-Theoretic IS Theory (DCEDM):** Building on *JMIS* scholarship on human-AI collaboration and delegation (Revilla et al., 2023; Tutun et al., 2024; Ansari and Ghasemaghaei, 2026) and neuro-cognitive warning habituation (Vance et al., 2018; Jensen et al., 2022), we formulate the **Dual-Cue Epistemic Defeasibility Model (DCEDM)**. Theorem 1 and Corollary 1 formally derive how narrative fluency ($\Delta F = 0.70$) shifts posterior compromise log-odds downward, compresses optimal primary telemetry verification ($V^*$), and inflates expected Compromise Miss Rates (CMR), while characterizing exact cost-ratio dominance boundaries ($\kappa^*(\lambda_F)$) and an interior optimal gating threshold ($\tau^*_{\text{LLR}}$) under unified endogenous warning habituation.

2. **Computational Cybersecurity Design Science Artifact (PITSA + Two-Stage DECS, DP1 to DP3):** Advancing the computational security design science stream in *JMIS* (Ebrahimi et al., 2020; Liu et al., 2020; Li and Chai, 2022; Ampel et al., 2024), we design the **Prompt-Isolated Telemetry Discrepancy Scorer (PITSA)** and **Two-Stage Defeasible Epistemic Cueing System (DECS)** around three generalizable socio-technical design principles (**DP1 to DP3**). Proposition 1 proves that enforcing an architectural projection $\Pi_{\text{struct}}$ ($\nabla_{\mathcal{A}_{\text{text}}} L(e) \equiv 0$) guarantees 100% invariance to text-channel prompt injection while providing white-box additive per-pillar forensic attribution ($k^*(e)$) in 2.12 to 4.31 $\mu$s per event on a single CPU core.

3. **Multi-Corpus Zero-Leakage Empirical Evaluation ($N = 2,238,434$ Real Telemetry Records):** Across four public cybersecurity corpora (`OTRF Security-Datasets`,

$N = 43,723$; `SALAD-SOC`, $N = 416,781$; `LMD-2023`, $N = 1,777,930$; and `CISA KEV`, $N = 1,741$ CVEs; 671,531 held-out test records), we demonstrate that five unpartitioned classifiers spanning lexical, dense semantic neural (SVD-48 + MLP(64, 32)), and contextual subword language model architectures suffer statistically significant recall collapse under both graded appended ITPI ($m \in \{1, 2, 4, 8\}$) and full-override ITPI (85.81% $\rightarrow$ 25.16% for Logistic Regression and 65.81% $\rightarrow$ 0.00% for Dense Neural MLP on OTRF; 77.46% to 84.93% clean recall dropping to 2.92% to 36.05% on SALAD-SOC; McNemar's $\chi^2 \geq 86.01$ on OTRF and $\chi^2 \geq 8,361.46$ on SALAD-SOC, all $p < 0.0001$). In contrast, Partitioned PITSA suffers 0.00 pp degradation (AUROC = 0.9807 on OTRF, 0.9380 on SALAD-SOC, and 0.9963 on LMD-2023).

4. **Trace-Driven Governance Sensitivity and Explicit Scope Boundaries:** Trace-driven evaluation over held-out LLR streams establishes that Selective DECS ($\Pi_4$) reduces warning banner volume by 95.92% (OTRF) and 79.87% (SALAD-SOC), cuts expected operational loss by 30.16% to 60.77% on enterprise SIEM alerts ($\kappa^* = 1.68$), and achieves an interior U-shaped habituation optimum at $\tau^*_{\text{LLR}} = 1.0$. We also transparently quantify the exact adaptive Living-off-the-Land (LOLBin) boundary where 5-pillar mimicry ($\rho_{\text{mask}} = 1.0$) defeats single-event structural scoring.

## Submission Declarations and Double-Blind Compliance

This manuscript is original, has not been published previously, and is not under consideration at any other journal or conference. The main manuscript (`paper.tex` / `paper.pdf`) is fully anonymized for double-blind review, with all code and multi-corpus artifacts openly accessible at https://anonymous.4open.science/r/epistemic-fluency-trap/. This study involves zero human participants, surveys, or interviews. The author declares no conflicts of interest and no external funding.

Thank you very much for considering our manuscript for *Journal of Management Information Systems*.

Sincerely,

**Tran Duc Le**
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